CSE 4615: Wireless Networks

B.Sc. Engg. CSE 6th Semester
Summer Semester, 2023-2024
Quiz: 1

Duration: 30 minutes Total Marks: 10 Date: 24.06.2025

Answer all three (3) questions. Figures inside the boxes in the right margin indicate marks of each question.

Student ID: ……..

1. Using a suitable real-world analogy and labeled diagrams, explain how ***Orthogonal Frequency Division Multiplexing (OFDM)*** enables *high spectral efficiency* and *interference-resistant* data transmission in modern wireless communication systems. 3

Sample Answer:

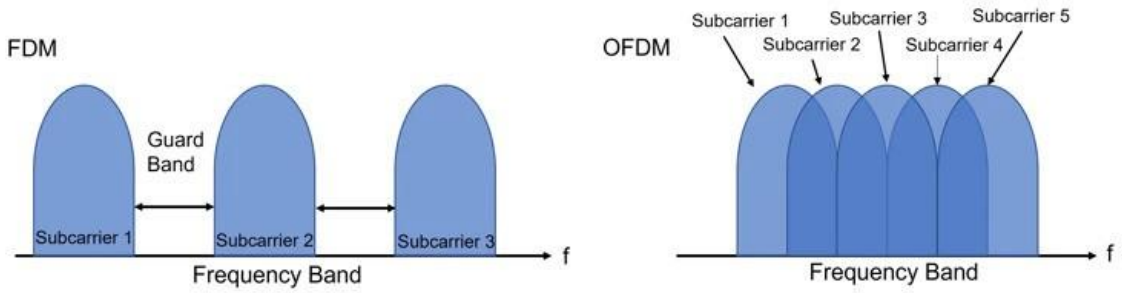
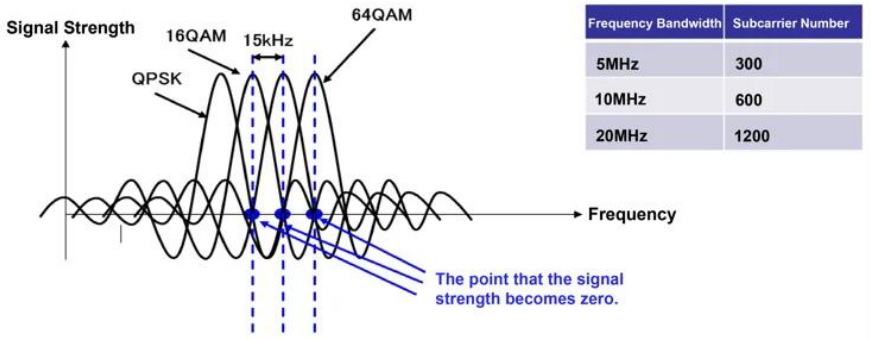


Figure: Conceptual diagram of multi-carrier transmission



Frequency Bandwidth	Subcarrier Number
5MHz	300
10MHz	600
20MHz	1200

Figure: Subcarrier arrangement in OFDM

Analogy for OFDM
FDM (Frequency Division Multiplexing):
Imagine several people in a room trying to talk at once. To avoid disturbing each other, they stand far apart and speak in separate corners. This avoids overlapping but wastes a lot of space (spectrum).

OFDM (Orthogonal Frequency Division Multiplexing):
Now imagine the same people sitting closer together, each speaking at a different pitch (frequency). Even though their voices partially overlap, the pitches are carefully chosen so they do not interfere with one another (orthogonal).

As a result, more people can talk clearly in the same space, achieving high spectral efficiency without confusion.

Orthogonal Frequency Division Multiplexing (OFDM) improves wireless communication by:

- **High Spectral Efficiency:** Subcarriers are placed very close together but remain orthogonal, avoiding interference and reducing wasted bandwidth.
- **Interference Resistance:** By dividing data into many low-rate streams over multiple subcarriers, OFDM minimizes the impact of multipath fading and inter-symbol interference.
- **Modern Use:** This makes OFDM the core of Wi-Fi, LTE, and 5G, supporting fast and reliable data transmission.

2. What environmental and technical factors contribute to a higher ***Bit Error Rate (BER)*** in wireless communication compared to wired communication? Briefly explain how each factor increases *BER*, providing clear and detailed arguments. 4

Sample Answer:

Environmental and Technical Factors Contributing to Higher BER in Wireless Communication:

1. **Signal Attenuation**
   - Wireless signals weaken as they travel farther or pass through obstacles (walls, particles, buildings).
   - Lower signal strength reduces the Signal-to-Noise Ratio (SNR), making it harder for the receiver to correctly detect bits, thus increasing BER.
2. **Noise and Interference**
   - Wireless channels are exposed to interference from other electronic devices and overlapping transmissions.
   - This mixing of unwanted signals with the desired one introduces errors in the received data.
3. **Multipath Fading and Environmental Obstructions**
   - Reflections from buildings, vehicles, or terrain cause the same signal to arrive at different times.
   - These delayed copies distort the original signal (fading or echoes), increasing bit errors.
4. **Doppler Shift (Mobility)**
   - Movement of either the transmitter or receiver causes frequency shifts.
   - This makes synchronization and signal decoding more difficult, raising the BER in mobile environments.
5. **Broadcast Limitations in Wireless Links**
   - Unlike wired links, where transmission is point-to-point and shielded, wireless broadcasts share the medium.
   - Limited control over collisions and acknowledgments increases the likelihood of errors.

3. Explain the following aspects of ***Wireless Mesh Networks (WMNs)*** in brief: 3
   i. How *WMNs* operate
   ii. Key challenges in their design and deployment
   iii. Real-world application examples

Sample Answer:

i. How WMNs Operate

- **Multi-hop Communication:** Data is forwarded through multiple intermediate wireless nodes before reaching the destination.
- **Infrastructure-based Access:** Mesh routers (fixed) form the backbone, while mesh clients (mobile or end devices) connect through them.
- **Self-Organization:** Nodes can automatically discover neighbors and establish routes dynamically.
- **Resilience:** If one path fails, data is rerouted via alternative mesh nodes.

ii. Key Challenges in Design and Deployment

- **Routing Complexity:** Efficient multi-hop routing under dynamic topologies is difficult.
- **Interference & Capacity:** Shared wireless medium causes interference and reduced throughput in dense deployments.
- **Scalability:** Ensuring reliable performance as the network size grows.
- **Mobility Support:** Maintaining seamless connectivity for mobile clients.
- **Security:** Vulnerability to attacks due to distributed and wireless nature.

iii. Real-World Application Examples

- **Community/City-wide Wi-Fi:** Municipal broadband networks for public internet access.
- **Disaster Recovery Networks:** Quick deployment in areas lacking infrastructure.
- **Military & Tactical Networks:** Secure, robust communications in the field.
- **Industrial IoT & Smart Cities:** Connecting sensors, utility meters, and surveillance systems.